

No	Item
1	What problem does semantic HTML solve, and when would you choose <section> vs <article> vs <div>?
2	Contrast <head> vs <header>; list 5 common elements that belong in <head>
3	Explain the meta viewport tag and its impact on mobile layouts.
4	Define the CSS cascade and the roles of origin, specificity, order, and layers.
5	Compare inherit, initial, unset, and revert (incl. revert-layer). Give a practical use for each
6	Describe specificity in (A,B,C) terms. Why is #id .btn:hover winning over .card .btn?
7	What are basic selector types, combinators, and pseudo-classes/elements you must master first? (Give examples.)
8	Define the box model and box-sizing; why does border-box simplify layout math?
9	What is margin collapsing and when does it not occur (e.g., flex & grid contexts)?
10	Explain overflow values (visible, hidden, scroll, auto, clip) and when each is appropriate.
11	Compare display modes relevant to layout: block, inline, inline-block, flow-root, flex, grid. (High-level.)
12	Positioning: static vs relative vs absolute vs fixed vs sticky; define “containing block” and stacking context implications.
13	Flexbox vs Grid—when to choose each; one-dimensional vs two-dimensional layout.
14	Responsive design toolbox: media queries vs container queries—differences and ideal use cases.
15	CSS units you’ll actually use: px, %, em, rem, vw, vh, plus math with calc() / clamp()
16	Images: purpose of alt, when to use <picture> vs srcset/sizes on , and how browsers pick a source.
17	Async vs defer on <script>—how loading & execution order change and why it matters.

18	Code: <pre> ''' <button class="btn primary">Buy</button> <style> .btn { color:#222; } .primary { color:#0a0; } button.btn.primary:hover { color:#070; } #cta .btn { color:#c00; } </style> ''' </pre> Question: On hover, what color renders and why (cite the selector that wins)? Why?
19	Code: <pre> ''' <article class="card"><p>Hello world</p></article> <style> .card { color:#333; } p { color:#666; } strong { color:inherit; } </style> ''' </pre> Question: What color is and why (inheritance rules)?
20	Code: <pre> ''' <div class="wrap"> <h2>Title</h2> <p>Para</p> </div> <style> h2 { margin:24px 0; } p { margin:16px 0; } .wrap { background:#f7f7f7; } </style> ''' </pre> Question: What's the vertical gap between h2 and p? Between .wrap top edge and h2? Explain collapse rules.

21	Code: <pre> ''' <section class="box"><div class="badge"></div></section> <style> .box { position:relative; width:200px; height:100px; } .badge { position:absolute; top:50%; left:50%; width:20px; height:20px; } </style> ''' </pre> Question: Where does .badge land and what is its containing block? Why?
22	Code: <pre> ''' ''' </pre> Question: On a 1000px viewport where the image layout width is ~500px, which file loads and why?
23	Code: <pre> ''' <script src="a.js" async></script> <script src="b.js" defer></script> ''' </pre> Question: Which runs first and why? What guarantees do you get?
1	What are the official ways to include Bootstrap (CDN, bundles vs separate files, package managers), and when would you choose each path?
2	Explain the roles of .container, .container-fluid, and responsive containers. When do max-widths kick in?
3	Describe Bootstrap's 12-column Grid: how rows, columns, and breakpoints interact (xs→xxl)
4	How do gutters work? Compare .g-*, .gx-*, .gy-*, and when to use g-0.
5	What are spacing utilities (m*, p*, responsive variants) and the mental model behind their shorthand?
6	Explain display utilities (.d-none, .d-md-block, etc.) for responsive show/hide behaviors.
7	Outline the Flex utilities (container + alignment + ordering) and when you'd favor flex utilities over writing custom CSS.
8	When should you reach for CSS Grid in Bootstrap (the optional CSS-Grid layout) vs the default Flexbox grid?
9	What does Reboot do and why is it shipped by default? Give two concrete fixes it provides.

10	What's the difference between color utilities (e.g., .text-*, .bg-*) and customizing colors via Sass variables?
11	Explain the Utilities API (how utilities are generated; why teams extend utilities instead of adding ad-hoc CSS).
12	Why are data attributes namespaced to data-bs-* in v5, and what practical benefits does this bring?
13	Contrast data-API initialization vs programmatic JS for components; when is each preferred?
14	Outline Forms fundamentals (form controls, layout, accessibility) and how validation styles integrate with native HTML5 validation
15	What does the Ratio helper do, and why is it useful for embeds and media?
16	What problem does .visually-hidden solve and how does -focusable change behavior?
17	What are Bootstrap Icons and how are they related (or not) to core Bootstrap?
18	Code: <pre> ... <div class="row"> <div class="col-12 col-md-6">A</div> <div class="col-12 col-md-6">B</div> </div> ... </pre> Question: How do A and B lay out at 576px, 768px, and 1200px viewports? Explain using breakpoint rules.
19	Code: <pre> ... <div class="row"> <div class="col-md-4 offset-md-2">One</div> <div class="col-md-4">Two</div> </div> ... </pre> Question: On ≥768px, where does One appear horizontally and why? What happens <768px?
20	Code: <pre> ... <div class="d-flex"> <div class="order-2 order-md-0">Left?</div> <div class="order-1">Right?</div> </div> ... </pre> Question: Which item renders first on mobile vs ≥md, and why?

21	Code: <pre> ''' <div class="row g-0"> <div class="col">X</div> <div class="col">Y</div> </div> ''' </pre> Question: Will there be spacing between X and Y? How would .gx-3 change it?
22	Code: <pre> ''' <p class="d-none d-md-block d-xxl-none">Hello</p> ''' </pre> Question: At which viewport ranges is “Hello” visible? Explain the combination.
23	Code: <pre> ''' <div class="row"> <div class="col-6"> <div class="col-6">Nested</div> </div> </div> ''' </pre> Question: Why is the nested .col-6 misaligned? What wrapper is missing?
24	List all primitive types in ES5 and how typeof reports each (including the famous typeof null)
25	Explain loose vs. strict equality (== vs ===) and when coercion happens. Give two surprising coercion examples and how to avoid them.
26	What is type coercion vs type conversion in JavaScript? Why does it matter for comparisons and arithmetic?
27	When and why should you always pass a radix to parseInt? What are the implications for strings like "08"?
28	How do NaN, isNaN() and numeric coercion interact? Why is Number.isNaN preferable when available?
29	What’s an IIFE?
30	What does "use strict" change compared to sloppy mode (e.g., implicit globals)? How do you enable it for a file or a single function?
31	Explain Array ES5 method: forEach VS map
32	Explain Array ES5 method: filter
33	Explain Array ES5 method: sort
34	Explain Array ES5 method: reduce
35	DOM model 101: what’s a Node vs an Element? Give examples of non-element nodes

36	How do you create, insert, and remove DOM nodes with createElement, appendChild, and insertBefore? When is insertBefore handy?
37	Contrast children vs childNodes (and what each includes).
38	querySelectorAll returns a NodeList. Is it live or static? Contrast with HTMLCollection. Why can forEach work on one but not the other in older code?
39	Compare textContent, innerText, and innerHTML. When is innerHTML risky?
40	What does classList provide over className? List common operations.
41	Define DOMContentLoaded vs the page load event. Why do many scripts bind to DOMContentLoaded?
42	Events: define capturing, target, and bubbling phases; what is the default?
43	What do event.preventDefault(), event.stopPropagation(), and event.stopImmediatePropagation() each do?
44	Explain event delegation and why it's efficient for dynamic lists. How do event.target and bubbling make it work?
45	Why is addEventListener generally preferred to inline handlers / on* properties? How do you remove a listener correctly?
46	Code: <pre> ''' console.log(0 == false, 0 === false, '0' == 0, '0' === 0); ''' </pre> Question: Predict the four outputs and explain why.
47	Code: <pre> ''' var a = null, b = [1,2], c = function(){}; console.log(typeof a, typeof b, typeof c); ''' </pre> Question: What prints and why is one of them historically surprising?
48	Code: <pre> ''' console.log(msg); var msg = 'hi'; say(); function say(){ console.log('ok'); } ''' </pre> Question: Explain the two outputs and which lines are hoisted.
49	Code: <pre> ''' run(); var run = function(){ console.log('go'); }; ''' </pre> Question: What happens and why? Contrast with a function declaration.

50	Code: <pre> ... function who(){ console.log(this.tag); } var tag = 'global'; var obj = { tag: 'obj', who: who }; who(); obj.who(); who.call({ tag: 'call' }); ... </pre> Question: Explain all three outputs under non-strict mode in a browser.
51	Code: <pre> ... function sum(a,b){ return a+b; } console.log(sum.call(null, 2, 3)); console.log(sum.apply(null, [2,3])); ... </pre> Question: Why do both log the same result, and what's the difference in how args are passed?
52	Code: <pre> ... var counter = { n: 0, inc: function(){ this.n++; } }; var fn = counter.inc; fn(); console.log(counter.n); ... </pre> Question: Why is n unchanged, and how would bind fix it?
53	Code: <pre> ... function makeAdder(x){ return function(y){ return x + y; }; } var add5 = makeAdder(5); console.log(add5(3)); ... </pre> Question: Why does this print 8? Describe what's "remembered."

54	Code: <pre> ''' (function(){ var secret = 42; })(); console.log(typeof secret); ''' </pre> Question: Why is secret not accessible here?
55	Code: <pre> ''' function pick(){ var args = Array.prototype.slice.call(arguments, 1); return args.join('-'); } console.log(pick('a','b','c')); ''' </pre> Question: What prints and why use slice.call?
56	Code: <pre> ''' var nums = [1,2,3]; var out1 = nums.forEach(function(n){ return n*2; }); var out2 = nums.map(function(n){ return n*2; }); console.log(out1, out2); ''' </pre> Question: Why is out1 undefined but out2 an array?
57	Code: <pre> ''' var ul = document.createElement('ul'); var li = document.createElement('li'); li.textContent = 'Item'; ul.appendChild(li); ''' </pre> Question: What appears in the UI and why is this approach safer than using innerHTML with user input?
58	Code: <pre> ''' var p = document.getElementById('p'); var b = document.createElement('b'); b.textContent = 'Hey'; p.parentNode.insertBefore(b, p); ''' </pre> Question: Where does land and why use insertBefore here?

59	Code: <pre> ... var a = document.querySelectorAll('.item'); // NodeList var b = document.getElementsByClassName('item'); // HTMLCollection ... </pre> Question: Which is live, which is static, and what's the practical impact when DOM changes later?
60	Code: <pre> ... var el = document.getElementById('box'); console.log(el.children.length, el.childNodes.length); ... </pre> Question: Why might these numbers differ? Give examples of nodes counted in childNodes.
61	Code: <pre> ... var d = document.getElementById('out'); d.textContent = 'Hello'; ... </pre> Question: What renders and why might this be preferable for untrusted strings?
62	Code: <pre> ... document.addEventListener('DOMContentLoaded', function(){ console.log('ready'); }); ... </pre> Question: When does this fire relative to images and stylesheets, and why is it ideal for DOM wiring?
63	Code: <pre> ... parent.addEventListener('click', log, true); // capture child.addEventListener('click', log); // bubble ... </pre> Question: Which handler runs first on a child click, and why? How would you reverse it?

64	Code: <pre> ''' link.addEventListener('click', function(e){ e.preventDefault(); // e.stopPropagation(); }); ''' </pre> Question: What behavior changes when you uncomment stopPropagation()?
65	Code: <pre> ''' function onClick(){ /* ... */ } btn.addEventListener('click', onClick); // later btn.removeEventListener('click', onClick); ''' </pre> Question: Why wouldn't an inline anonymous function work for removal here?

Front-End Essentia

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Day 2: Bootstrap & JavaScr

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